INCH-POUND
MIL-DTL-32015/1A(AS)
6 May 2004
MIL-DTL-32015/1(AS)
15 October 1999

DETAIL SPECIFICATION SHEET

MICROPHONE, LINEAR, BOOM TYPE M-177/AIC

This specification is approved for use by the Naval Air Systems Command, Department of Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-32015.

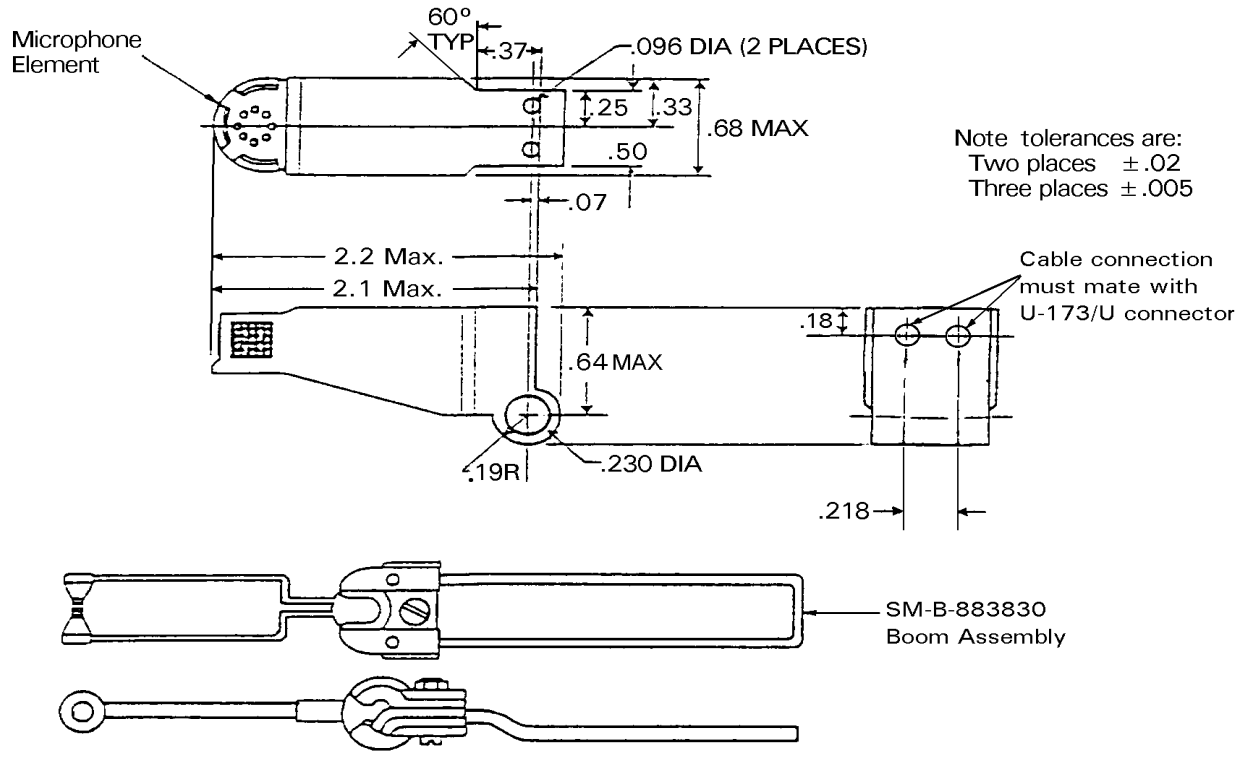

*FIGURE 1. Boom microphone assembly

AMSC N/A FSC 5965

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APPLICABLE DOCUMENTS:

MIL-DTL-22442/36 Cable Assembly, Aircraft Audio

REQUIREMENTS:

*DESIGN and CONSTRUCTION: The microphone design shall be within the envelope dimensions shown. The shape may differ.

COMPONENT PARTS: The microphone assembly shall consist of the following:

1 Microphone element
1 SM-B-883830 Boom assembly
1 M22442/36-X Cable assembly

*PART or IDENTIFYING NUMBERS (PIN):

M32015/1-1	With MIL-DTL-22442/36-3 13 inch cable assembly.
M32015/1-2	With MIL-DTL-22442/36-4 16 inch cable assembly.
M32015/1-3	With MIL-DTL-22442/36-1 6-inch cable assembly.
M32015/1-4	With MIL-DTL-22442/36-6 23-inch cable assembly.
M32015/1-5	Microphone element only

WEIGHT: 62g max. for the assembly.

COLOR: Semigloss, lusterless, black.

WIND SCREEN: Each microphone shall be supplied with a wind screen firmly attached.

*INPUT CURRENT: 20 mA max. at 27.5 VDC when connected as shown on figure 2 with switch SW1 in the open position

IMPEDANCE: The electrical impedance of the microphone assembly shall be such that the unit shall meet the sensitivity and frequency response to a close sound source while connected to a 100 ohm ±1 percent load impedance.

*SENSITIVITY: 64 mV to 130 mV at 104 dB SPL with 27.5 VDC supply voltage, and shall not vary by more than 1.0 dB when the supply voltage is varied between 22 to 30 volts.

SENSITIVITY AT REDUCED BIAS VOLTAGE: Shall not vary by more than 12 dB at 104 dB SPL from the 27.5 VDC established value when the supply voltage is varied between 12 and 22 volts.

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SENSITIVITY AT ALTITUDE: ±3 dB of sensitivity at ground level while at a simulated altitude of 15,000 ft above sea level.

FREQUENCY RESPONSE AT GROUND LEVEL: See figure 3.

FREQUENCY RESPONSE AT ALTITUDE: Within ±5 dB of the frequency response at sea level, while at a simulated altitude of 15,000 feet above sea level.

NOISE IMMUNITY: See figure 4.

*LINEARITY: The microphone signal level shall be a linear function of acoustic input SPL from 74 to 120 dB, when connected in accordance with figure 2.

*HARMONIC DISTORTION: The harmonic distortion in the output of the microphone shall be no greater than 2 percent up to 114 dB SPL and no greater than 5 percent from 114 to 120 dB SPL.

GAIN STABILIZATION: Within 100 mS when supply voltage is between 23 and 30 VDC. Within 150 mS when supply voltage is between 12 and 22 VDC.

OVERVOLTAGE PROTECTION: 15 minutes operation in the test circuit shown on figure 2 with 27.5 volts applied while the 330 ohm resistor is shorted.

EXPLOSIVE ATMOSPHERE: N/A

*RAPID DECOMPRESSION: As specified in the general specification.

ENVIRONMENTAL STRESS SCREENING (ESS): 24 hours of temperature cycling between -20 and +85 °C, with each cycle consisting of 1 hour at each temperature and the change between temperatures occurring at a minimum rate of 5 °C per minute. The microphone assemblies shall be functional tested prior to each cold to hot transition and upon completion of the temperature cycling.

MARKING:

M-177/AIC
M32015/1-X
MANUFACTURER'S NAME OR CAGE CODE
CONTRACT NUMBER
DATE CODE
SERIAL NUMBER

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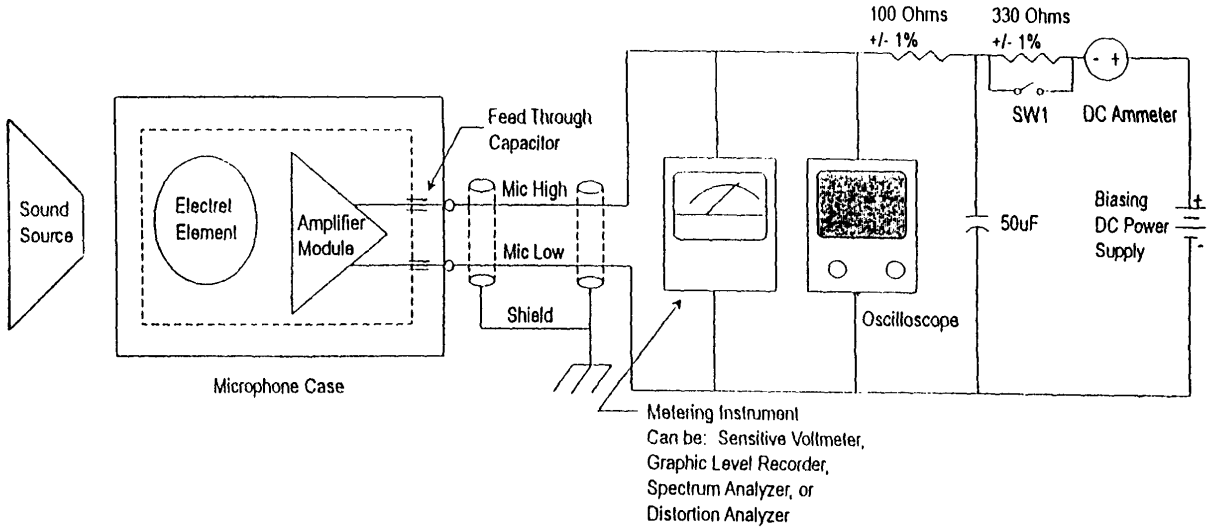
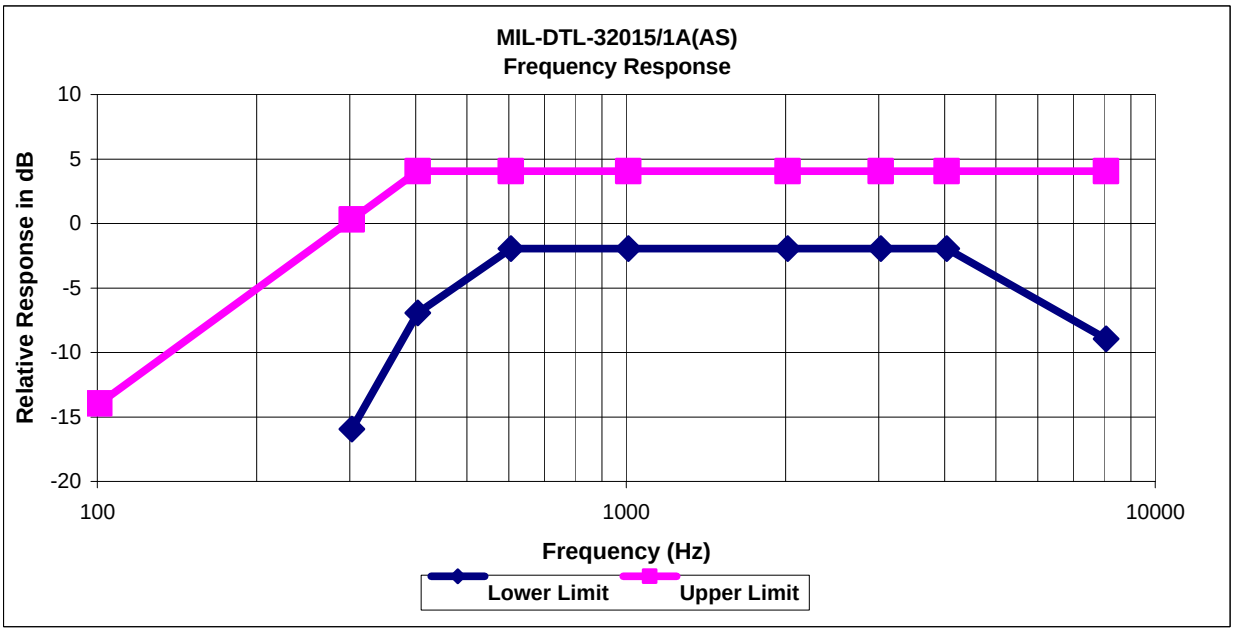


FIGURE 2. Test circuit schematic



* FIGURE 3. Frequency response limit curve.

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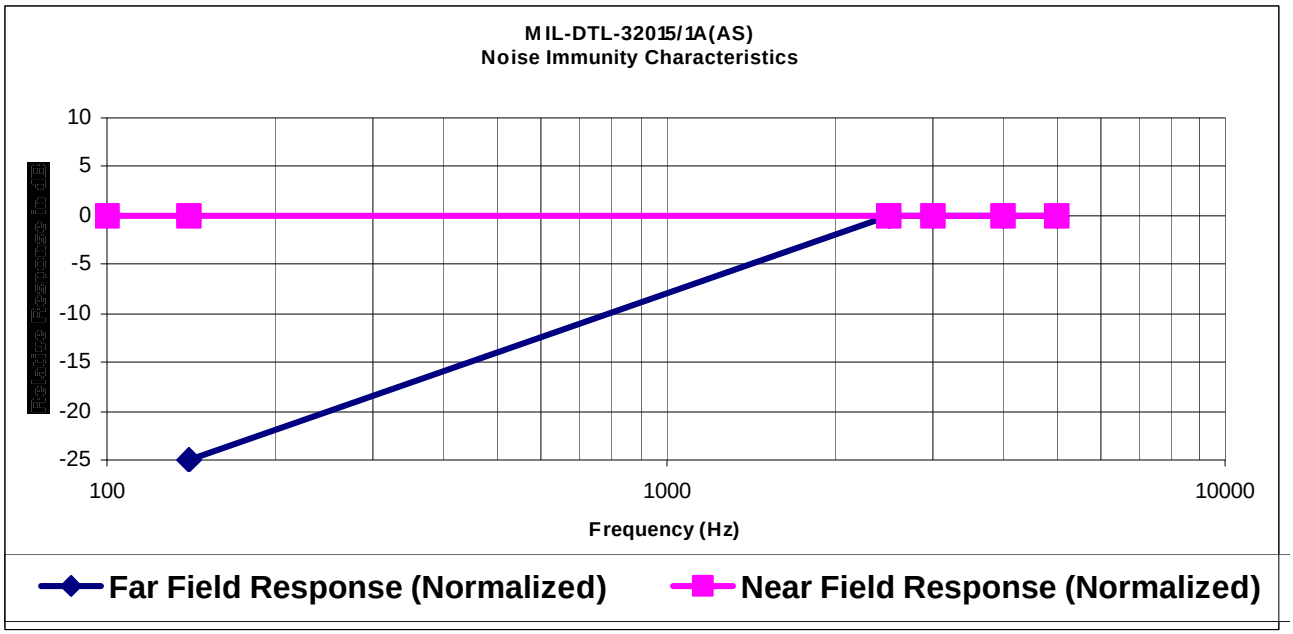

* FIGURE 4. Noise immunity characteristics

INTENDED USE: The microphone, M32015/1-X (see PIN), is a noise cancelling, dc-powered, high-gain microphone designed for boom mount use on Navy fixed- and rotary- wing helmets. The microphone is intended for use as a part of the airborne communication system which will provide communication of high intelligibility under the extreme noise conditions encountered in military aircraft. The microphone is designed to work with the AIC-14 intercommunication system, specifically control box C-2645D.

NOTE: The margins of this amendment are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Preparing activity:
Navy - AS
(Project 5965-0388)

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